Generating Functions

$$G(x) = \sum_{k=0}^{\infty} a_k x^k \longrightarrow$$ A convenient trick to record a bunch of numbers.

I Dice Example

- Consider a standard 6-sided dice.
- Probability to roll a 1 = p_1

$2 = p_2$
$3 = p_3$
$4 = p_4$
$5 = p_5$
$6 = p_6$

How can we create a function that contains all the information about its probabilities?

In the eyes of a generating function, just write the sequence with incrementing powers of some variable x

$$G(x) = p_1 x + p_2 x^2 + p_3 x^3 + p_4 x^4 + p_5 x^5 + p_6 x^6 \qquad (1)$$

But how do we get the properties from (1)?

↳ What math operations can we <u>apply</u> to $G(x)$ to get p_1? p_2? ...

$$G'(x) = p_1 + 2p_2 x + 3p_3 x^2 + \dots$$

↳ set $x = 0$ and $G'(0) = p_1$

$$G''(0) = 2p_2 \Rightarrow \tfrac{1}{2} G''(0) = p_2$$

$\Rightarrow$ To get the value: $\boxed{p_k = \frac{1}{k!} \frac{d^k G(x)}{dx^k}\Big|_{x=0}}$

II So Why do we care?

Derivative of $G(x)$

A) Consider the dice problem where each coefficient is the probability of the dice roll. Suppose we have

$$G'(x) = p_1 + 2p_2x + 3p_3x^2 + 4p_4x^3 + 5p_5x^4 + 6p_6x^5$$

↳ Let $x=1$

$$G(1) = p_1 + 2p_2 + 3p_3 + 4p_4 + 5p_5 + 6p_6$$

This is the expected value (average) of the dice rolls!

$$E(x) = \sum_{x \in A} x\, p(x)$$

B) Multiplying Generating Functions

$$G(x)^2 = p_1p_1x^2 + (p_1p_2 + p_2p_1)x^3 + (p_1p_3 + p_2p_2 + p_3p_1)x^4 + \dots$$

Imaginge rolling 2 die now. What is the probability of rolling a 2?

↳ $p_1p_1x^2$ $(1+1=2)$

Probability of rolling 4?

$(p_1p_3 + p_2p_2 + p_3p_1)x^4$ $(1+3;\ 2+2,\ 3+1) = 4$

$\Rightarrow$ <u>$G(x) \times G(x)$ creates another $G(x)$</u>